

PROJECT TASK-2

Combinational and Sequential RTL Design, FSMs, Counters, Shift Registers, Parameterized Verilog Design, Timing Concepts, and Advanced RTL Simulation

Industry-Oriented Internship Program on VLSI Design and Semiconductor Engineering

1. Introduction

Task-2 advances the RTL skills developed in Task-1 by focusing on sequential logic, counters, shift registers, parameterized RTL, finite state machines, timing analysis, advanced testbench development, and engineering documentation.

2. Tools Used

Operating System	Ubuntu Linux through WSL
HDL	SystemVerilog / Verilog HDL
Simulator	Icarus Verilog
Waveform Viewer	GTKWave
Editor	Visual Studio Code
Version Control	Git and GitHub

3. Sequential RTL Design

Implemented and simulated:

- D Flip-Flop with synchronous reset
- 8-bit Register with synchronous reset
- 8-bit Shift Register with serial input

The sequential designs use clocked RTL and non-blocking assignments. Waveforms were generated with Icarus Verilog and inspected in GTKWave.

4. Counter Design

Implemented and verified:

- 4-bit Up Counter
- 4-bit Down Counter
- Mod-10 Counter
- Ring Counter
- Johnson Counter

The counter sequences were verified through simulation and waveform inspection. Documentation includes block diagrams, truth/state tables, and timing information.

5. Parameterized RTL Design

Reusable parameterized modules were implemented for:

- N-bit Register
- N-bit Counter

- N-bit Shift Register

Multiple widths were tested through simulation, demonstrating reusable RTL rather than hard-coded widths.

6. Traffic Light FSM

A Moore finite state machine was implemented for a Traffic Light Controller. The controller contains RED, GREEN, and YELLOW states and performs automatic state transitions. Reset places the controller in the RED state.

Current State	Output	Next State
RED	Red = 1	GREEN
GREEN	Green = 1	YELLOW
YELLOW	Yellow = 1	RED

Automatic testbench checks were used to verify GREEN, YELLOW, RED, and GREEN states. No ERROR messages were produced during the completed FSM verification.

7. Timing Analysis

The testbenches use a 10 ns clock period generated with `always #5 clk = ~clk;`, corresponding to a 100 MHz clock. Timing documentation covers rising-edge behavior, synchronous reset, setup time, hold time, and the relationship between clock transitions and sequential outputs.

8. Advanced Testbench Development

A reusable testbench template was created with automated clock generation, reset generation, a reusable reset task, waveform dumping, and simulation completion reporting. The reusable testbench was successfully executed with Icarus Verilog.

9. Documentation Produced

The Week-2 project contains RTL block diagrams, FSM diagrams, truth tables, timing documentation, RTL notes, simulation-results documentation, waveform screenshots, and a README. These materials organize the design and verification evidence required by the task.

10. Verification Summary

Design Area	Verification Result
D Flip-Flop	Simulation completed; waveform inspected
8-bit Register	Simulation completed; waveform inspected
Shift Register	Simulation completed; waveform inspected
Counters	Simulation completed; sequences verified
Parameterized Modules	Multiple widths verified
Traffic Light FSM	Automatic checks PASS
Reusable Testbench	Execution PASS

11. Repository Organization

The Week-2 work is organized under `week-2/sequential-logic/` with separate directories for RTL, verification, waveforms, documentation, and advanced testbenches. The work has been committed and pushed to the GitHub repository.

12. Conclusion

Task-2 work demonstrates practical RTL design and simulation skills across sequential circuits, counters, parameterized hardware, and a Moore FSM. The completed designs were simulated using Icarus Verilog, waveforms were inspected using GTKWave, and supporting engineering documentation was prepared. The remaining submission items are the final project PDF and daily progress log, which are included as separate deliverables.

Prepared for Task-2 submission